RESEARCH-TO-PRODUCT IMPLEMENTATION GUIDE

Coffee White Stem Borer Image Detection AI

Complete end-to-end team handbook

Beginner-friendly instructions for collecting verified Arabica coffee photographs, preparing labels, training a robust image classifier, handling bright sunlight, evaluating unseen estates, and deploying a cautious farmer-facing system.

Prepared for team review • Version 1.0 • 11 November 2026

Starting position	What it means
315 reported infected-plant photographs	Reported by field team; files have NOT been supplied or audited in this document. Number of distinct plants unknown.
Healthy / negative photographs	Quantity and inspection status not confirmed; must be collected or verified.
Model currently trained?	No. This is an executable implementation plan, not a trained or validated CWSB model.
Primary scientific reference	Bhandarkar et al. (2019), ISPRS Archives, DOI 10.5194/isprs-archives-XLII-3-W6-443-2019.

Read this first

The aim is a visual screening system for suspect CWSB symptoms, not a guarantee that photographs can see living larvae hidden inside intact wood. The application must allow “inconclusive” and must not instruct destructive treatment based only on an AI result.

This handbook is arranged in the exact order a new team should execute the work. Every phase states: the goal, simple explanation, action steps, files to produce, and a pass/fail gate.

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NOTE | The document is intentionally detailed. Start with Phase 0 and do not skip the data-audit gate.

01 Fundamentals and project definition

1.1 What is CWSB?

Coffee White Stem Borer (CWSB), Xylotrechus quadripes, is a beetle pest of Arabica coffee. Larvae develop inside the stem; externally visible signs can include bark ridges and, in advanced cases, yellowing and wilting. An exit hole may show that an adult already emerged. These visible signs do not prove that a live larva is presently inside the photographed stem. Sources: Coffee Board planter guidance [R2], Coffee Board pest guidance [R3].

1.2 What is machine learning?

Machine learning means showing a program many examples with reliable answers. Here, an example is a photograph plus an expert-confirmed plant-level label. The program learns numeric image patterns and later predicts a label for an unseen photograph. It does not memorize plant biology in the way an agronomist does.

1.3 Define the product goal carefully

V1 goal: when a farmer submits a usable coffee-plant and main-stem photograph, identify whether the visible appearance resembles expert-confirmed CWSB cases sufficiently to advise inspection. Output is an advisory, not a confirmed diagnosis. The model is an image classifier; it is NOT a weather-based emergence forecast.

Will do in Version 1	Will NOT claim in Version 1
Classify visible evidence in photos	Detect invisible early larvae inside intact stems
Accept/decline low-quality inputs	Infer infestation from any yellow leaf
Provide inspect / no visible signs / inconclusive	Guarantee pest absence or presence
Record farmer and expert feedback	Prescribe uprooting or insecticides automatically
Measure performance on new farms	Reuse 2019 paper accuracy as our accuracy

1.4 Define labels before collection

Plant-level ground-truth labels are not identical to image quality or AI outputs. Keep them separate in metadata.

Ground-truth label	Who assigns it	Required support
CWSB_CONFIRMED	Trained agronomist/entomologist	Document observed evidence and diagnosis method; record internal larva/tunnel if known.
CWSB_NOT_DETECTED	Trained field inspector	Document actual stem inspection; this is not a guarantee of biological absence.
OTHER_STRESS_NO_CWSB_FOUND	Expert	Record alternative issue and documented CWSB inspection.
UNRESOLVED	Reviewer	Evidence inadequate, mixed symptoms, or diagnosis unverified. Exclude from supervised labels until resolved.

NOTE | For a first binary model, map CWSB_CONFIRMED to positive. Use inspected negative plants as negatives; include other-stress negatives where confidently inspected. Never label "unhealthy leaf" as positive without CWSB confirmation.

1.5 Define success

A proof-of-concept success is a trained, reproducible classifier with a transparent, independent evaluation—not merely high training accuracy. A field pilot can proceed only after the team agrees to numerical acceptance thresholds and reviews missed confirmed cases. Example aspirational criteria: CWSB recall >= 90% on clearly visible confirmed cases and acceptable false alerts; these are project targets to validate, not scientific guarantees.

02 Scientific reference and limits

2.1 What the uploaded 2019 study actually implemented

Bhandarkar et al. (2019) described 2,425 images from coffee-growing regions of Karnataka and image-processing, transfer-learning, and autoencoder experiments. The paper reports an Inception-v3-based transfer-learning average accuracy of 85.5%; a 512 x 512 processed-image experiment reached 87%. It also tested separate weather-based regression models, which are NOT our image-system target. Its stated counts—400 healthy + 800 infested + 400 discarded/dead—total 1,600, not 2,425; four classes are referenced without a fully reproducible class specification. The raw training images are not included in the PDF. Reference [R1], pp. 443–451.

2.2 Lessons to retain

- Start with a pretrained CNN instead of training millions of image parameters from zero.
- Test on natural field photographs, not just laboratory images.
- Use direct stem evidence instead of relying only on whole-plant colour.
- Reproduce the evaluation method on our own data; do not borrow the paper's reported accuracy.
- Treat paper ambiguities (class definitions, original image list, split protocol) as unknowns, not facts.

2.3 Evidence hierarchy

Evidence level	Examples	Use
A — Confirmed plant evidence	Expert field record with observed CWSB evidence	Main target labels
B — Strong but incomplete	Stem ridges / exit hole photo, not confirmed as current CWSB	Review queue; not automatically positive
C — Non-specific symptoms	Yellow foliage, general wilting	Other-stress features, not positive ground truth
D — Public supplementary datasets	Coffee leaf images with labels for other conditions	Supplement negative diversity only; validate provenance

03 Find and acquire appropriate data

3.1 The non-negotiable data sources

Source	Available evidence	Role / limitation
Field team: reported 315 infected photos	User-reported count, not yet audited	Potential positive set; must verify labels and unique plant count.
Coffee Board / CCRI / 2019 study authors	2019 publication reports positive and healthy photographs	Request originals, labels, site IDs, consent and permitted-use terms. Dataset link NOT verified.
Local Indian Arabica estates	Team can capture whole plant and stem	Primary external performance data. Required.
JMuBEN2 (Kenya)	Healthy Arabica leaf images; some augmented	Supplementary healthy foliage, NOT verified healthy stem/CWSB-negative plants.
BRACOT (Brazil)	300 tree-portion photographs, leaf stresses	Background / other-stress diversity; no verified CWSB positives.
BRACOL (Brazil)	Coffee leaf disease / healthy images	Other-stress negatives, not CWSB-specific.

Direct links are in Section 16. Research paper PDF license does not automatically license the underlying photographs. Confirm copyright, consent and commercial-use permission per dataset.

3.2 Request the original 2019 data

1. Identify corresponding authors and CCRI point of contact from [R1].
2. Request every original image (not only published figure thumbnails), with image ID, expert label, date, estate, plant ID and stage definitions.
3. Ask whether images were augmented and whether different images show the same plant.
4. Ask whether data were included in their train/test splits and what images were excluded.
5. Request a written license / research or commercial usage grant and redistribution limits.
6. If not available, plan the pilot around field collection instead of searching indefinitely.

3.3 The minimum collection matrix

Plan sufficient independent plants, not merely images. For a pilot, aim for about 300 verified CWSB plants and 300 inspected non-CWSB plants, with three photo views each. This is a work-planning target, NOT a guaranteed sample size for accuracy. Add plants with other diseases / nutrient stress to test false alarms. Let the learning curve and field validation determine further collection.

Variety / state	Capture views	Lighting balance
Confirmed Arabica CWSB	Whole plant, full stem, close-up of affected stem	Sun, partial shade, diffuse light; multiple phones
Arabica inspected, no CWSB detected	Whole plant, full stem, matching bark close-up	Same lighting, phones and estate types as positives
Arabica other-stress, CWSB not detected	Whole plant, stem, symptomatic leaf if present	Match positive collection methods
Unresolved plants	Collect for review; NOT train as known	Store separately

Variety / state	Capture views	Lighting balance
	class	

04 Audit the 315 reported infected photographs

4.1 Do not begin training until this audit is complete

The only confirmed fact presently available is the field report of 315 images. Those image files have not been attached for inspection. We do not know whether all show a visible stem, whether they are duplicates, how many unique plants they represent, or how each diagnosis was confirmed.

7. Copy originals into an immutable folder: data/raw/field_315/. Never edit in place.
8. Create one metadata row per image and record its source/owner.
9. Count readable image files; flag corrupt or unsupported files.
10. Compute SHA-256 exact hashes; mark exact duplicates without deleting originals.
11. Create perceptual-hash or embedding-based candidate near-duplicate groups. Human reviewer confirms true duplicates.
12. Assign plant_id and estate_id, including when multiple photographs show the same stem.
13. Record view: whole_plant, main_stem, closeup, leaf_only, unusable.
14. Mark severe blur, blown highlights, deep shadows and obscured stem.
15. Request expert review of the reported “infected” label, with evidence grade.
16. Create an audit report: usable images, unique plants, confirmation status, lighting balance and collection gaps.

4.2 Deliverable format: image audit summary

Audit metric	Value at project start	After audit
Incoming images	315 reported	Count actual files
Exact duplicates	Unknown	Verified count
Near duplicates	Unknown	Confirmed groups
Unique plants	Unknown	Plant IDs
Expert-confirmed CWSB	Unknown	Positive plant count
Main-stem visible	Unknown	Usable stem-photo count
Direct-sun photos	Unknown	Count
Poor-quality images	Unknown	Counts by failure reason

4.3 Audit decision

PASS: original data backed up; every retained image has plant_id, estate_id, view_type, provenance and verified or unresolved diagnosis. FAIL: file count or labels unclear; continue cleaning and field verification. Never inflate “315 images” into “315 independently confirmed plants.”

05 Field capture SOP: what farmers and field teams must do

5.1 Prepare the plant record

The collector first creates a plant ID. An expert or field technician documents whether the stem was inspected. The operator does not assign “healthy” merely because foliage is green. Record species Arabica, estate, block, approximate plant age if known, operator, date/time, device and image conditions.

5.2 Required photographs

17. Whole plant: hold the camera approximately level; include whole canopy, avoid extreme backlight.
18. Main stem: show lower and middle main stem, bark and thick primaries where visible; focus carefully.
19. Close-up: capture ridges, cracks, suspected holes or a comparable bark area for negatives; include enough surrounding bark to recognize context.
20. Optional extra view: opposing side of stem, different angle, or reference scale. Do not replace the main-stem photo with a leaf-only photo.

5.3 Lighting instructions

Prefer bright but diffuse daylight. If direct sun creates saturated white regions, move position or shade the photographed area without altering plant evidence. Avoid flash glare against wet bark. Keep real examples of direct sun, patchy shade and dark shade for robustness tests. Record lighting condition instead of trying to force every photo to look identical.

5.4 Quality checklist for each photo

- Subject is Arabica coffee and main diagnostic region is visible.
- Image not severely blurred; texture can be inspected at native resolution.
- Exposure has not destroyed relevant stem detail.
- Stem is not blocked by leaves / hands / labels.
- Image is saved at original resolution; no social-media compression if possible.
- Plant ID and view type are linked correctly to the photo.
- Photograph is taken before any destructive inspection if possible; record any later confirmation separately.

5.5 Field safety and label integrity

Do not cut or uproot plants solely to obtain training labels. Follow qualified agronomic guidance, estate permissions and appropriate safety rules. Expert review can use already-authorized pest management inspections. Consent and site privacy must be handled before sharing estate photos.

06 Annotation: teach the model the correct answer

6.1 Two annotation levels

Plant-level label: the expert-confirmed outcome of the plant. Image-level view/quality tags: which symptom is visible and whether the photograph is useful. A plant can be confirmed infested even if one

of its photographs does not show visible CWSB signs. Do not falsely claim that every image of that plant displays a visible lesion.

Annotation field	Allowed example	Meaning
plant_diagnosis	CWSB_CONFIRMED	Verified at the plant level
diagnosis_method	larva_observed / expert_stem_inspection	How diagnosis was established
view_type	whole_plant / main_stem / closeup / leaf_only	What the photo shows
visible_symptom	ridge / exit_hole / wilt / none / unclear	Observation, not automatically diagnosis
lighting	direct_sun / partial_shade / diffuse / low_light	Capture condition
quality	usable / recapture / review	Whether image can be used
label_confidence	high / medium / low	Review certainty

6.2 How to annotate in CVAT

21. Create a CVAT project called CWSB_Arabica_Photos.
22. Import image batches with stable filenames and the metadata CSV kept separately.
23. Create image tags for view_type, quality, lighting and visible symptoms.
24. Use boxes or polygons only when you need spatial localization of ridges / holes in a future detector.
25. Assign an expert reviewer to confirm plant-level status in the metadata system.
26. Have a second reviewer examine ambiguous / difficult positives and negatives.
27. Record every corrected label in a review log, preserving the original and reason.
28. Export annotations with version number and keep originals unchanged.

6.3 Difficult cases: decision rule

If foliage is yellow but stem evidence is missing, label image-level symptom as “yellowing”; plant diagnosis remains unresolved unless expert evidence exists. If an exit hole is visible, record “exit_hole”; do not automatically claim a living larva. If only leaf images are available, keep them for an alternative-stress task, not as primary evidence for visible stem signs.

6.4 Annotation pass gate

PASS when at least two reviewers agree on ambiguous examples, all supervised examples have documented provenance, and class definitions are frozen in dataset version 1.0. FAIL when positives are labelled solely by colour or when the same plant has contradictory labels without explanation.

07 Data storage, privacy, provenance and versioning

7.1 Keep original data immutable

```
CWSB_AI/
 data/
  raw/field_315/          # untouched originals
  raw/healthy_field/      # untouched originals
```

```
  raw/external_sources/     # keep source license
  metadata/images.csv
  metadata/plants.csv
  annotations/v1/
  splits/v1/{train,val,test}.csv
src/{audit.py,train.py,evaluate.py,predict.py}
configs/
models/
reports/
api/
docs/
```

7.2 Source registry

For each external image collection, store source URL, author, DOI, version, license, restrictions, whether images are augmented, original source identity, download date, and whether commercial model training is permitted. If the rights are unclear, do not place the images in the training dataset until permission is clarified.

7.3 Data protection

Remove unnecessary EXIF location data before public sharing; store farm/plant coordinates separately with access controls. Define retention periods, farmer consent, restricted staff access, backup and a deletion process for photographs. Capture the model's privacy responsibilities in the app policy.

7.4 Version everything

Name artifacts clearly: dataset_v1.0, labels_v1.0, split_v1.0, train_config_v1, model_v1, evaluation_v1. Archive every training configuration and fixed random seed. Once the test dataset is selected, do not repeatedly use it to choose model settings.

08 Prevent data leakage: build reliable splits

8.1 What is leakage?

If the same plant is photographed three times and photos appear in both training and testing, a neural network may recognize the plant or background instead of general CWSB evidence. Public augmented images may also be near-copies of one original. Both cause inflated scores.

8.2 Recommended split rules

29. Group all photographs of the same plant under a single plant_id.
30. If plant_id is missing, investigate same-session stems and backgrounds before splitting.
31. Keep original and all augmented copies in the same split; augmentation belongs only to training.
32. For first experiments, use approximately 70% of plants for train, 15% validation, 15% test, stratifying by label and estate if feasible.
33. Reserve separate entire estates as an external test; if too few estates exist, report that limitation.
34. Do not include test images in preprocessing selection, threshold calibration or model iteration.

8.3 Example

A photographed infested plant P023 may have whole_plant.jpg, stem.jpg and closeup.jpg taken in three light conditions. ALL nine images are assigned to one group, and the group goes wholly into train or validation or test.

8.4 Leakage checks

Write a script that asserts train/val/test plant_id sets are disjoint; check SHA-256 overlaps; run cross-split near-duplicate alerts; manually review suspicious clusters. Record group counts, not only photo counts.

09 Install tools and train Model V1

9.1 Tools explained simply

Tool	Beginner meaning	Why needed
Python	Programming language	Training and prediction scripts
VS Code	Code editor	Manage project files and commands
TensorFlow/Keras	Deep-learning framework	Inception-v3 and training
OpenCV	Image-processing library	Quality and lighting experiments
pandas	Table/CSV library	Read image and plant metadata
scikit-learn	Evaluation toolkit	Splits, precision, recall, confusion matrix
CVAT	Annotation website/app	Image tags and optional bounding boxes
MLflow	Experiment tracker	Keep training runs and metrics reproducible
FastAPI	Python web API framework	Connect model to the mobile app

9.2 Windows installation walkthrough

35. Install a currently TensorFlow-supported Python release from python.org. Select “Add Python to PATH” where offered.
36. Install VS Code and its Python extension.
37. Open the CWSB_AI folder in VS Code. Choose Terminal > New Terminal.
38. Create an isolated environment with the following commands.
39. If CPU training is too slow, use a supported cloud GPU or Linux/WSL2 CUDA stack; do not assume native Windows GPU works for all TensorFlow versions.

```
python --version
python -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
python -m pip install tensorflow pandas numpy pillow opencv-python scikit-learn matplotlib
```

9.3 What the first CNN does

Use pretrained Inception-v3 with a new two-class classification head. Inception-v3 learns general edges, shapes and texture from earlier image training. Our labels teach the final layers to distinguish verified CWSB-associated images from inspected negative examples. Choose a second three-class model only if other-stress labels are trustworthy and numerous enough.

9.4 Processing each image

Read original RGB image; verify readable; apply aspect-ratio-aware crop/pad to 299 x 299; use the correct Inception-v3 preprocessing convention; augment only training examples. DO NOT apply inconsistent normalization twice. Do not apply identical random brightness changes to validation or test photographs.

9.5 Two-stage training

Stage	What to do	Initial experiment (to tune)
1. Frozen feature extractor	Freeze pretrained backbone; train only classification head.	Adam, LR ~1e-3, batch 16/32, early stopping
2. Fine-tuning	Unfreeze limited upper layers; keep BatchNorm handling stable.	LR ~1e-5, small number of epochs, monitor val PR-AUC/recall
3. Calibration	Select alert / abstain thresholds on validation.	Calibration curve, sensitivity analysis
4. Final test	Evaluate ONCE on untouched held-out and external estates.	Freeze model and thresholds first

All settings are proposed starting points, not guaranteed optimal hyperparameters. Preserve epoch logs, learning curves, selected checkpoint and label mapping.

9.6 Core training pseudocode

1. load metadata and verified labels
2. group data by plant_id / estate_id
3. create train, validation and test manifests
4. load InceptionV3(weights="imagenet", include_top=False)
5. freeze backbone, attach pooling + dropout + 2-class head
6. train with balanced sampling / class weighting if needed
7. evaluate validation metrics; fine-tune upper layers
8. calibrate decision + abstention thresholds on validation
9. lock trained weights and evaluate untouched test
10. save model, config, class map and evaluation report

9.7 Common beginner mistakes

- Using a random split by file rather than by plant.
- Training on public leaf images labelled “healthy” while positives are Indian stem photos; network learns image style/site rather than pest evidence.
- Applying random augmentations to the test set.
- Treating cross-entropy loss or training accuracy as field accuracy.
- Copying published 85.5% as our expected accuracy.

- Claiming output confidence is calibrated without testing.
- Changing test thresholds repeatedly until the result looks better.

10 Sunlight resilience: filters, training and rejection

10.1 Why sunlight matters

Direct sun causes strong highlights, shadows and white-balance shifts. Dark bark can become underexposed, while reflective leaves can lose texture. A CNN may learn background and lighting differences unless both classes are photographed under similar conditions.

10.2 Do NOT rely on one “magic filter”

First collect real lighting diversity. Second, train with moderate photometric augmentation. Third, test optional white balance, luminance-only CLAHE or gamma correction. Fourth, reject unusable inputs. Once sensor highlights are clipped, a filter cannot faithfully recover missing diagnostic texture.

10.3 Quality gate algorithm

Check	How to implement	What to do
Too blurry	Laplacian variance as a diagnostic; calibrate per resolution and crop	Ask for sharper stem photo
Excess clipped highlights	Fraction of pixels near maximum intensity in stem ROI	Ask for new angle / diffuse light
Dark diagnostic region	Percent low-luminance pixels in stem ROI	Ask for improved illumination
Wrong subject / no stem	Human pilot checks, later detector	Ask for main-stem photo
Too distant / tiny lesion	Stem occupies too few pixels	Request closer photo

There is no universal numeric blur or glare threshold; derive thresholds from examples of acceptable and unacceptable field images. Evaluate each subgroup and avoid rejecting all direct-sun images unfairly.

10.4 Lighting experiment plan

40. Create a paired test set: same confirmed plants in sun, shade and diffuse light, where practical.
41. Train A: originals + realistic brightness/contrast/white-balance augmentation.
42. Train B: optional luminance correction + same augmentation.
43. Freeze each pipeline; compare the same untouched test set by lighting and device.
44. Select the pipeline with best sensitivity, false alerts, calibration and image retention, not just best overall accuracy.

10.5 Initial augmentation ranges to test

Transform	Candidate range	Warning
Brightness	~ +/- 20%	Do not erase real yellowing
Contrast	~ +/- 15%	Can amplify bark noise
Saturation	~ +/- 10%	Large shifts destroy colour clues

Transform	Candidate range	Warning
Rotation	~ +/- 15 degrees	Keep relevant stem features visible
Crop/zoom	Mild	Never crop all evidence away
Blur/noise	Occasional mild variation	Avoid making every training image unreadable

11 Evaluation: prove it works

11.1 Different metrics answer different questions

Metric	Plain-English meaning	Why it is needed
TP	Confirmed CWSB correctly flagged	True detections
FN	Confirmed CWSB not flagged	Potentially costly misses
FP	No CWSB found but AI flags CWSB	Unnecessary field inspections
TN	No CWSB found and AI does not flag	Correct negatives
Recall = TP/(TP+FN)	Share of confirmed cases caught	Key sensitivity metric
Precision = TP/(TP+FP)	Share of alerts that are correct	Controls nuisance alerts
F1	Balance of precision and recall	Summary metric
PR-AUC	Precision-recall across thresholds	Useful with rare positives
Calibration	Whether predicted scores match observed rates	Interpretability of probability-like scores
Abstain rate	Share reported inconclusive	Measures usability and cautiousness

11.2 Test matrix

Test slice	Mandatory output
Overall held-out plants	Confusion matrix, recall, precision, PR-AUC, CIs
New estates	Same metrics, count of plants per estate
Direct sun / partial shade / diffuse	False-negative rate and rejection rate per lighting group
Whole plant vs stem photos	Performance for each view
Early externally visible vs advanced	Recall by verified severity if labelled
Other stresses	False alarms for rust, drought, nutrient symptoms
Different phones	Sensitivity and image failure by device
Missing / poor-quality views	How often model abstains

11.3 Final safety gate

Before pilot launch, freeze the target population, label protocol, alert threshold, inconclusive threshold and required metrics. Suggested project goal (to be approved): high recall for visible, expert-confirmed cases, with manageable false alarms; report uncertainty intervals and site-specific results. Do NOT claim early internal detection when none is externally visible. A trained classifier is not a substitute for field inspection.

11.4 Model error-review meeting

Review every false negative visually with an entomologist: Was the stem visible? Was the ground truth correct? Was the symptom faint or hidden? Was the example from a new camera/estate? Do not quietly delete difficult cases from evaluation.

12 Mobile application and backend architecture

12.1 Farmer journey

45. Farmer opens app and selects “Check White Stem Borer”.
46. App explains that it screens visible signs and requests whole-plant and main-stem views.
47. App checks consent, camera permissions and image quality.
48. Upload images securely; assign plant/session identifier.
49. Backend performs preprocessing identical to training, then CNN inference.
50. Decision layer returns: CWSB signs suspected / no visible CWSB signs / inconclusive.
51. App gives simple guidance: inspect main stem and request expert confirmation if suspected.
52. User may send feedback; expert-confirmed diagnosis is recorded separately.

12.2 Proposed backend

Phone app -> API gateway/authorization -> image storage -> quality service -> model inference -> calibrated decision rule -> response -> audit database -> optional expert review queue. Record model version, preprocessing version and threshold version with every result. Use HTTPS, access controls and rate limits.

12.3 Example API contract (illustrative)

```
POST /v1/cwsb/predict
multipart/form-data:
  plant_id=F01-P023
  whole_plant=<file>
  main_stem=<file>
  closeup=<optional file>

Response (illustrative):
{
  "status": "inconclusive",
  "reason_codes": ["stem_not_visible"],
  "advice": "Please take a clear main-stem photo.",
  "model_version": "cwsb_v1"
}
```

12.4 Do not expose uncalibrated scores as certainty

A raw softmax score is a model number, not necessarily a trustworthy probability of CWSB. If showing a percentage to farmers, calibrate it and validate against field data. Consider simpler qualitative categories for initial pilot release.

12.5 Deployment choices

Begin with server-side prediction using FastAPI + a versioned Keras/PyTorch model. After evaluation, compare a smaller MobileNet-style model for offline processing if necessary. Measure inference latency on actual network conditions and define upload size limits without aggressively losing stem texture.

13 Operations: improve after first pilot

13.1 Monitoring dashboard

Track daily predictions, image rejection reasons, percentage of inconclusive cases, site/device/light distribution, expert confirmation rate, false alerts and misses. Privacy-preserving logs should include model and dataset version.

13.2 Retraining loop

53. Collect farmer examples in a review queue; keep user reports distinct from expert-verified outcomes.
54. Prioritize examples where model and experts disagree or lighting is difficult.
55. Have the entomologist verify labels; add to a new dataset version.
56. Retrain a candidate model; compare against frozen benchmark tests and a new external estate.
57. Release only if the candidate meets gates; preserve rollback to the prior model.

13.3 Operational controls

Require approvals for dataset changes; set a model owner; keep a rollback plan; publish a clear limitation statement; audit any advice wording. Keep images and diagnosis metadata secured, and remove data on request under the project's data policy.

14 Team, schedule and project gates

14.1 Responsibility matrix

Role	Key ownership	Deliverable
Project owner	Scope, budget, decisions	Signed requirements and release gate
Entomologist / agronomist	Case definitions, ground truth, error review	Approved label SOP and confirmed cases
Field lead	Plant identification, photo collection	Originals, plant metadata, lighting coverage
Dataset/annotation lead	QC, CVAT workflow, licenses	Audited versioned dataset
ML engineer	Training, calibration, evaluation	Reproducible model and model card
Backend engineer	Inference API, auth, storage	Versioned and tested API
App engineer	Photo guide and farmer response	Pilot capture and advisory experience
QA tester	New-farm usability tests	Field test report and release recommendation

14.2 Example plan (depends on data access)

Weeks	Focus	Gate
1–2	Class definitions, 315-image audit, permissions	G0: Labels and plant IDs recoverable
2–6	Collect inspected negatives, other stresses, lighting diversity	G1: Dataset sufficiently verified
4–7	Annotate and split; near-duplicate review	G2: Leakage and quality audit pass
7–9	Inception-v3 baseline, learning curves	G3: Baseline reproducible
9–11	Sunlight correction ablation, thresholds	G4: Validation accepted
11–13	Hold-out estate testing; expert review	G5: External test decision
13–16	API, app integration and pilot	G6: Restricted pilot launch

This timeline is illustrative and assumes access to enough verified plants. Lack of CWSB-positive diversity or expert inspection may take longer. Do not shorten independent validation to meet a schedule.

14.3 Project risk register

Risk	Impact	Mitigation
315 photos are only a few unique plants	High leakage / poor generalization	Group by plant and collect more distinct plants
Positive labels lack expert evidence	Model learns noisy targets	Adjudicate and quarantine unresolved records
Healthy photos from another country look different	Shortcut learning by geography/device	Match local negative collection conditions
Severe glare / shade hides bark details	Missed CWSB	Quality gate, diverse collection, abstention
Other diseases cause yellow leaves	False positives	Hard negatives; prioritize stem evidence
Data licensing unclear	Legal/commercial risk	Written permission and source registry
Single-estate validation	Overoptimistic accuracy	Independent estate holdout
App treats AI as diagnosis	Harmful farm decisions	Cautious language and expert referral

14.4 Procurement / resource needs

Minimum: smartphones with decent cameras, protected image storage, 1–2 trained field collectors, access to an agronomist, one Python-capable ML developer, annotation capacity, and a computer/cloud GPU for training. Cost and final GPU choice depend on the actual dataset size and deployment volume; avoid buying large GPU hardware before a working pilot.

15 Ready-to-use templates

15.1 images.csv schema with example rows

```
image_id,plant_id,estate_id,filename,view_type,lighting,quality,source_id
```

```
IMG001,FARM01_P001,FARM01,data/raw/field_315/IMG001.jpg,main_stem,direct_sun,review,FIELD_2026
IMG002,FARM01_P001,FARM01,data/raw/field_315/IMG002.jpg,closeup,partial_shade,usable,FIELD_2026
```

15.2 plants.csv schema with example rows

```
plant_id,estate_id,species,plant_diagnosis,diagnosis_method,verified_by,verified_date
FARM01_P001,FARM01,Coffea arabica,CWSB_CONFIRMED,expert_stem_inspection,REVIEWER01,2026-11-11
```

15.3 Model experiment record

Field	Example
Experiment ID	INCEPTION_V1_AUGMENT_A
Dataset / split	dataset_v1 / split_v1
Pretrained weights	ImageNet Inception-v3
Image preprocessing	299x299 + correct preprocessor
Trainable layers	Head then top blocks
Augmentation	Brightness 0.8-1.2, etc.
Selection metric	Validation CWSB recall + precision and PR-AUC
Final report	Held-out confusion matrix + lighting and estate slices

15.4 Acceptance checklist before team sign-off

[ ] The 315 reported images are audited and unique plants counted.

[ ] Every training image has a source and permitted use.

[ ] All positive labels have appropriate expert evidence.

[ ] Inspected negative examples include matching field conditions.

[ ] Other-stress examples have been considered.

[ ] Train / validation / test contain disjoint plants and derivative groups.

[ ] External estate or documented alternative external test exists.

[ ] Sunlight and shade results are reported separately.

[ ] Model thresholds and inconclusive rules are validation-based.

[ ] Model never presents an image-only result as a confirmed living larva.

[ ] Mobile app provides safe recapture and expert-inspection advice.

[ ] All artifacts are versioned and a rollback procedure exists.

15.5 First-week action list (start here)

58. Day 1: Receive all 315 image files; create backups; obtain estate and plant IDs from field team.
59. Day 2: Verify file count, duplicates and usable stem photos.
60. Day 3: Ask agronomist to verify CWSB labels and identify unresolved images.
61. Day 4: Create local inspected healthy image collection protocol; identify participating estates.

62. Day 5: Contact CCRI / paper authors; request original study dataset and license.
63. Day 6–7: Produce audited dataset_v0 summary, class gaps and field collection schedule.

16 References, dataset links and beginner glossary

16.1 Sources and what each supports

[R1] Bhandarkar et al., 2019, ISPRS Archives — Original paper: image collection, transfer learning, model performance and limitations.
https://doi.org/10.5194/isprs-archives-XLII-3-W6-443-2019

[R2] Coffee Board India, Planter guidance — Official pest symptoms and management; not an image dataset.
https://coffeeboard.gov.in/planter.aspx

[R3] Coffee Board, Management of Major Coffee Insect Pests (2021) — CWSB biology, visible symptoms and management context.
https://coffeeboard.gov.in/Publications/January2021_final.pdf

[R4] JMuBEN2 Mendeley Data — Healthy Arabica leaf images and leaf-miner examples; not CWSB stem labels.
https://data.mendeley.com/datasets/tgv3zb82nd/1

[R5] JMuBEN dataset article, Data in Brief — Description and provenance of coffee leaf disease data.
https://doi.org/10.1016/j.dib.2021.107142

[R6] BRACOT Mendeley Data — Arabica tree portions and other biotic stresses; not CWSB labels.
https://data.mendeley.com/datasets/pmkbyjpf6k

[R7] BRACOL Mendeley Data — Coffee leaf disease/healthy examples; not CWSB labels.
https://data.mendeley.com/datasets/yy2k5y8mxg/1

[R8] Keras Inception-v3 — Framework documentation for the chosen baseline.
https://keras.io/api/applications/inceptionv3/

[R9] TensorFlow install — Current Python and platform setup requirements.
https://www.tensorflow.org/install

[R10] CVAT documentation — Annotation setup and exports.
https://docs.cvat.ai/

[R11] scikit-learn metrics — Evaluation metrics and validation tools.
https://scikit-learn.org/stable/modules/model_evaluation.html

[R12] FastAPI documentation — Prediction API development.
https://fastapi.tiangolo.com/

Source discipline: [R1] is a peer-reviewed published study. [R2] and [R3] are Indian Coffee Board guidance. [R4]–[R7] are supplementary datasets. Engineering choices, quantity targets, schedules and accept/reject policies in this handbook are proposals that require team testing and approval.

16.2 Glossary for first-time readers

Term	Plain-English meaning
AI	Software that uses learned patterns to make predictions.
CNN	Convolutional neural network; a model commonly used for images.
ImageNet	Large general image dataset used to pretrain popular CNNs.
Transfer learning	Starting from a pretrained model and adapting it to our task.
Epoch	One full pass through the training examples.
Batch	A group of images processed together during training.
Label	An expert-provided answer linked to an image or plant.
Ground truth	Best available verified observation used to evaluate predictions.
Positive	Here, an expert-confirmed CWSB plant.
Negative	A plant inspected with no CWSB detected under the recorded protocol.
False negative	A confirmed CWSB example that the model fails to flag.
False positive	A negative example wrongly flagged as CWSB.
Data leakage	Information from test examples entering training or model decisions.
Augmentation	Realistic training-time variation of existing images.
CLAHE	Local contrast enhancement; optional, not a cure for blown highlights.
Calibration	Checking whether model scores reflect observed rates.
Abstention	Allowing the AI to decline a weak or unsuitable prediction.
Inference	Running a trained model on a new photograph.
API	Software interface used by a mobile app to request a prediction.
MLOps	Processes for versioning, monitoring, testing and updating models.

Document control

Prepared: 11 November 2026. Audience: field collection team, agronomy experts, data annotators, ML engineers, backend engineers and project owners. This handbook is a planning specification. It does not certify the 315 image files, provide a trained model, or verify future field performance.